Fall 2025

Operations Research (MTH601)

Assignment No. 1

Section In charge: Mr. Muhammad Nadeem

Total Marks: 40

Due date: 24th November 2025

Question 1: **Marks=10**

You are required to analyse the following project by drawing its network diagram, calculating the expected time and variance for each activity, identifying the critical path and the expected project completion time.

Activity	Optimistic (t_0)	Most Likely (t_m)	Pessimistic (t_p)
1–2	2	3	4
1–3	1	2	3
2–4	4	5	6
3–4	3	4	5
4–5	2	3	7

Question 2: **Marks=10**

A company uses 24,000 units of an item annually. The ordering cost is Rs. 60 per order and the holding cost is Rs. 1.50 per unit per year. Determine the Economic Order Quantity (EOQ), number of orders per year, and the time between two consecutive orders.

Question 3: **Marks=10**

A company produces an item at a rate of 50,000 units per year. The setup cost per production run is Rs. 200, and the holding cost is Rs. 2 per unit per year. The annual demand for the item is 20,000 units. Determine the optimum production quantity (Q*), maximum inventory level, and number of production runs per year.

Question 4: **Marks=10**

A company has an annual demand of 14,400 units for a component. The ordering cost is Rs. 450 per order, the holding cost is Rs. 1.50 per unit per year, and the shortage (back ordering) cost is Rs. 6.00 per unit per year. Calculate the Economic Order Quantity (EOQ) when planned shortages are allowed, the maximum optimal shortage quantity, and the time between orders (in months).